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Studierichting: Informatica

Oefeningenreeks 4A nr. 44.3

$$\int \frac{1}{x(x^2 + 1)} dx$$

$$\frac{1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{(x^2 + 1)}$$

$$\Rightarrow A = \frac{1}{x^2 + 1} = 1 \text{ met } x = 0$$

$$\Rightarrow Bi + C = \frac{1}{x} = \frac{1}{i} \text{ met } x = i$$

$$\Rightarrow Bi + C = \frac{1}{i} = \frac{i}{i \cdot i} = -i$$

$$\Rightarrow B = -1 \text{ en } C = 0$$

$$= \int \frac{1}{x(x^2 + 1)} dx = \int \frac{1}{x} dx + \int \frac{-x}{(x^2 + 1)} dx$$

$$= \ln|x| - \frac{1}{2} \ln|x^2 + 1| + c$$

References